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Review

Direct and indirect effects of visual pattern understanding on word reading in Chinese kindergarten children

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ABSTRACT

The present study investigated the direct effect of pattern understanding on Chinese word reading, and the indirect effects via metalinguistic awareness. Participants were 138 children (73 boys, mean age = 59.60 months at the first wave of assessment) recruited from four kindergartens in Hong Kong. They were assessed twice with an interval of 14 months. In the first wave of assessment, which took place in the spring term in the second year of kindergarten, children were assessed regarding pattern understanding, phonological awareness, morphological awareness, and orthographic awareness. Working memory, spatial visualization, and receptive vocabulary were also assessed as control variables. In the second wave, which took place at the end of the third kindergarten year, Chinese word reading was measured using two tasks, i.e. single-character word reading and double-character word reading. Results of a multivariate multiple regression showed that pattern understanding made a unique contribution to subsequent Chinese single- and double-character word reading independently of spatial visualization, working memory, vocabulary, and demographic variables. Results of the mediation model showed that the unique effects of pattern understanding were mediated by phonological awareness. The direct effect of pattern understanding remained significant on Chinese double-character reading. The findings highlight the importance of pattern understanding in Chinese word reading for Chinese kindergarten children.

1. Introduction

Patterning activities are frequently adopted in early childhood education classrooms in western countries (Sarama & Clements, 2009). Parents and teachers have reported that children frequently engage in pattern related activities. Such activities may promote children's development of pattern understanding (Pasnak et al., 2019), which refers to the ability to abstract rules in sequences and make predictions about subsequent elements. Findings from correlational and intervention studies (Burgoyne et al., 2019; Kidd et al., 2014; Liu et al., 2022; Pasnak et al., 2015) showed that pattern understanding contributes to word reading. However, the mechanism underlying the association remains unclear. Some researchers have argued that pattern understanding exerts an impact on reading skills as a proxy for general cognitive ability, e.g. fluid reasoning and executive function (Burgoyne et al., 2017). It has also been proposed that pattern understanding facilitates word reading, because pattern understanding helps with discovering the speech-print mapping pattern in writing systems (Manning et al., 1995). Having a grasp of why word reading and pattern understanding are related is important. It will shed light on the unique precursors of word reading and thus inform the practice of early literacy education.

Current empirical evidence is primarily from alphabetic languages. It remains unknown if pattern understanding contributes to word reading in a logographic writing system such as Chinese, in which the speech-print mapping is ambiguous and more complex. To fill this particular research gap, the present study investigated whether pattern understanding would predict subsequent Chinese word reading independently of other domain-general cognitive and language abilities, i.e., working memory, spatial visualization, and vocabulary. The mechanism of the prediction was further explored by examining the indirect effects via metalinguistic awareness skills that tap into three aspects of Chinese language and are involved in speech-print mapping. Testing this mechanism in a Chinese sample can inform the linguistic universality and specificity hypotheses in reading acquisition.

1.1. Pattern understanding

Pattern understanding has its origin in math education. Patterning

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understanding reflects looking for predictable sequences and identifying rules and regularities in them (Pasnak et al., 2015). In children's daily lives, sequences occur in many forms. Aspects such as color, shape, and spatial orientation can be altered to create sequences with underlying rules, in that the elements can be definitely predicted. Educators and clinicians in early childhood education settings have frequently adopted and endorsed a variety of sequences for their educational value (Hendershot et al., 2016; Sarama & Clements, 2009).

Pattern understanding as a skill in cognitive development was not been systematically investigated until the recent two decades. Patterns can be in various structures. Repeating patterns are those in which the elements alternate repeatedly (e.g., ABBABB). Young children gradually develop visual repeating pattern understanding in preschool and kindergarten (Papi et al., 2011; Rittle-Johnson et al., 2013; Sarama & Clements, 2009). By the end of preschool, many children are able to *duplicate* (make an exact replica of a model pattern), *extend* (continue an existing pattern by at least one unit of repeat), and *abstract* (recreate a model pattern using a different set of materials) repeating patterns (Papi et al., 2011; Rittle-Johnson et al., 2015; Sarama & Clements, 2009). Other than repeating patterns, elements in patterns can have quantitative or spatial relations. Growing patterns are those in which the number of elements in a unit systematically decreases or increases. Rotating patterns are those in which the units are the same object rotating by a fixed degree, e.g., 90°.

Patterns can also vary in terms of the nature of the element. Visual sequences are comprised of elements of objects from children's daily lives such as shapes, color blocks, and pictures, and contrast with alphanumeric sequences which are made of notational symbols such as letters and numbers (e.g., number patterns or number series; Zhang et al., 2017). The present study focuses on visual pattern understanding as a predictor for word reading, to exclude the confounding factor of symbolic knowledge, as letter and number naming have been shown to robustly predict Chinese word reading (Huo et al., 2021).

1.2. Pattern understanding, domain general abilities, and word reading

Some researchers attribute the association between word reading and pattern understanding to domain-general ability (Burgoyne et al., 2017). There is substantial evidence for the importance of fluid intelligence in word reading acquisition, especially in beginning readers in both alphabetic (Guerin et al., 2020) and non-alphabetic reading systems (Yang et al., 2019). The reason is that the task of learning to read is novel for young children and requires a lot of problem solving and reasoning at the early stage. It has also been shown that working memory is important in the beginning stage of learning to read (e.g., Chung & McBride-Chang, 2011), because holding phonological information, forms of visual symbols, and word meaning imposes heavily on memory capacity before word reading becomes automatized.

Pattern understanding has overlap with general cognitive abilities and executive functions, but it also taps into a unique aspect of cognitive ability. Pattern understanding is closely related to fluid intelligence, which is defined as the ability to think logically and solve problems in novel situations (Cattell, 1987). Pasnak et al. (2015) argue that pattern understanding is a form of fluid reasoning, which involves inductive and deductive thinking. In other words, pattern understanding is a construct narrower than fluid intelligence. Some components of fluid intelligence overlap with those of pattern understanding, such as relational and inductive thinking, both of which underlies rule abstraction (Pasnak et al., 2015). However, other components of fluid intelligence do not involve rule abstraction, and thus are distinct from pattern understanding. For example, spatial visualization, as a component in fluid intelligence (Keith et al., 2001; Salthouse et al., 2008), is often measured by the task of selecting visual shapes to form an intact picture. This process requires multi-step problem solving, the selection of a strategy, and cognitive flexibility, but there is no underlying rule to be detected in the items (Linn & Petersen, 1985). Moreover, even when assessing relational and inductive thinking, measures of pattern understanding are often simpler than those of fluid intelligence, as there is typically only one rule underlying a one-dimension sequence for the former, whereas the latter, primarily measured by matrices reasoning, involves reasoning about multiple rules underlying the columns and rows in the matrices (Carpenter et al., 1990). Moreover, the task of matrices reasoning provides very limited instructions, for the purpose of measuring knowledge-independent problem-solving skills (Cattell & Horn, 1978). In contrast, in patterning tasks, the concept about pattern/rules is explicitly explained to children, honing in on measuring pattern understanding rather than general problem solving. Finally, pattern understanding is found to be moderately related to working memory, another component of fluid intelligence (Miller et al., 2016). Specifically, children rely on working memory capacity to process complex relational information in pattern understanding.

Another perspective argues for specific linkage between pattern understanding and word reading. Words and sentences are sequences. Children identify relationships between spoken (temporal *sequences*) and written language (spatial *sequences*), and make predictions based on the rules underlying letter–sound relationships and word order ((Manning et al., 1995). Following this logic, at the word level, enhancing pattern understanding should improve children's knowledge and skill about sound-print mapping, which then facilitates word reading. Letter-sound knowledge and phonemic awareness, the ability to access and explicitly manipulate the smallest linguist unit in speech, enable one to apply the rules of letter–sound relationships, i.e., phonological decoding (Byrne & Fielding-Barnsley et al., 1989). There is some preliminary evidence for the linkage between pattern understanding and phonological skills. Intervention studies with first and second graders showed that patterning instruction had small effects on phonological decoding (see Burgoyne et al., 2017, for a review). Several correlational studies on pattern understanding measured word reading as well as phonological awareness and phonological decoding in kindergarten children (Burgoyne et al., 2019; Strauss et al., 2020). Other than the expected contributions of pattern understanding and phonological skills to word reading, the findings also showed moderate correlations between pattern understanding and phonological skills ($r = 0.2 \sim 0.5$). This tentatively suggests that pattern understanding could exert an effect on word reading via phonological skills.

Phonological skills are also important in Chinese word reading despite the fact that the Chinese writing system is logographic. Characters are the writing units in the Chinese orthography. Most Chinese characters are compound characters comprising two radicals, one cueing sound (phonetic radical) and the other cueing meaning (semantic radical). For example, the character 码 /ma3/ (meaning code) has the semantic radical 石and the phonetic radical 马/ma3/, the latter of which shares the same pronunciation with the compound character. Ho and Bryant (1997) found that Chinese children in grade 1 and 2 commonly made use of phonetic radicals in reading characters, which is reflected by the phonological errors they produced, i.e., overgeneralizing the pronunciation of the phonetic radicals in inconsistent characters. Moreover, the ability of using phonetic radical are predicted by phonological awareness (Ho & Bryant, 1997b).

In the Chinese writing system, phonological cues are not that reliable (Hoosain, 1993). The speech and print mapping features rules in other aspects beyond phonology, e.g., morphology and orthographic regularities. Admittedly, morphology and orthography are also important in opaque alphabetic writing systems such as English. But these two aspects are more important in Chinese than English (Perfetti et al., 2013; Ruan et al., 2018). Morphological awareness plays a particularly important role in multiple-character Chinese word reading (Li et al., 2014; Wang et al., 2014). Pattern understanding may be involved when children abstract rules in the aspects of morphology and orthography to predict the pronunciation of the characters indirectly. In terms of morphology, one case would be heteronyms, meaning that one character can correspond to multiple pronunciations. The pronunciation is only contextualized in multiple character words. For example, the character 重 is pronounced /zhong / in words like重要/important/, 重点/key points/, and 重量/weight/, representing morphemes that are semantically related; the character has another pronounciation /chong / in words like 重复/repetition/, 重叠/overlap/, and重来/do over/. Children identify the sound-print corresponding relationships of multiple-character words based on the morphological context. This ability is supported by compounding morphological awareness (MA), which means the access to the morphological structures and the ability to manipulate morphemes (McBride-Chang et al., 2007).

In addition to oral language skills, orthographic awareness (OA) is also important to word reading in young children. For young children who are beginning to learn to read, OA refers to the sensitivity to the configurative forms of Chinese characters. For example, kindergarten children are able to make distinctions between characters and foreign scripts and drawings (Zhang et al., 2017) and are able to judge whether a character is well written (Qian et al., 2015). It is found that four-year-old children's OA predicted subsequent Chinese word reading even after the baseline performance was controlled for (Huo et al., 2021; Yang et al., 2021). Pattern understanding might relate to OA, as knowing Chinese characters as the writing unit in the sequence of Chinese print helps to better process their visual forms. Previous studies (Liu et al., 2022) have reported moderate correlations between pattern understanding and children's OA ($r = 0.3$).

1.3. The present study

The present study is one of the pioneering efforts investigating the role of pattern understanding in Chinese word reading in preschool children. We adopted a short-term longitudinal design using early pattern understanding and other cognitive-linguistic skills to predict word reading skill a year later. The first question is to investigate the unique effect of pattern understanding on word reading. We included three visual patterning tasks, i.e., repeating patterns, growing patterns, and rotating patterns, to build a global measure of pattern understanding. Burgoyne et al. (2019) conducted a longitudinal study with 5-year--old children and found that pattern understanding uniquely predicted word reading one year later after controlling for a range of well-established literacy related skills and executive function. Yet, the influence of some components of fluid intelligence, such as spatial visualization, was not partialled out in that study. Therefore, we controlled for spatial visualization and working memory, two components of fluid intelligence distinct from pattern understanding. Vocabulary knowledge was also included as a control variable since it may serve as a proxy for general language development (Lee, 2011) and has a broad influence on metalinguistic awareness skills and word reading. Single-character and double-character word reading were included as two measures for Chinese word reading. Because previous studies have shown that different cognitive and developmental processes underlie single- and double-character word reading (see Li et al., 2014, for a review). The second question is whether metalinguistic awareness skills, i.e., PA, MA and OA, would mediate the effect of pattern understanding on Chinese word reading abilities. Specifically, we predicted that pattern understanding would contribute to phonological awareness which then predict word reading as suggested in previous studies (Burgoyne et al., 2019).We also predicted that single-character word reading relies more on orthographic processing, whereas MA makes an additional contribution to double-character word reading (Li et al., 2014; Wang et al., 2014).

2. Method

2.1. Participants

One hundred and thirty-eight children (73 boys), recruited from four kindergartens in Hong Kong, participated in the study (age in month: *mean* = 59.60, *sd* = 5.48). The four kindergartens were all non-profit-making ones. In all four kindergartens, the primary language of instruction was Cantonese. In Hong Kong, it is common that teachers explicitly teach word reading and writing in early childhood education settings. Demographic information was collected from the parents of 111 participants and was shown in Table 1. Most of the children were from families of middle class, because 55 % of the parents obtained a degree higher than high school diploma, and 61 % of the families had a monthly household income above the local median amount (HK$25,200 [US$3222]; Census and Statistics Department, 2020).

2.2. Procedure and sample attrition

Hong Kong children attend kindergarten for three years, usually from ages 3 to 6 years, before compulsory education starts at Grade 1. The first wave of data collection was conducted in April in the second year in kindergarten (Time 1 [T1]), and the second wave was conducted in June in the third year in kindergarten (Time 2 [T2]). Children's oral assents were sought at the beginning of the first assessment and parental consents were sought prior. In the first wave, 138 children participated in the assessment. Sixteen children missed one or multiple tasks due to absence for unknown reasons. Sixteen children did not perform the patterning tasks, one the working memory task, two the spatial relations task, eight the vocabulary task, two the morphological awareness task, and four the orthographic awareness task. The Hawkins test was performed to test the missingness of the data in the first time point. The result was non-significant, indicating that evidence was not enough to reject the assumption that the data was missing at random (MCAR) (Jamshidian et al., 2014). In the second wave, one hundred and twenty-nine children participated, and all completed the tasks of Chinese word reading. Attrition analysis was performed using logistic regression. The result indicated that a child dropping out in the second wave could not be predicted by T1 cognitive-linguistic skills (vocabulary, spatial visualization, working memory, pattern understanding, metalinguistic awareness skills) or any demographic variables (father educational attainment, mother educational attainment, age, family income). All the children were assessed individually in a quiet room in multiple sessions spread over several consecutive days. Each assessment session lasted less than 25 minutes to avoid fatigue.

Table 1
Parents' educational level and family income.

Monthly income (scored below from 1 to 20)	Mother	Father	Total
1–5.<5000 HKD ($640)	35 %	7.2 %	3 %
6–10. 5000–9999 HKD ($1280)	7.3 %	2 %	5 %
11. 10,000–19,999 HKD ($2560)	10.8 %	19.8 %	18 %
12. 20,000–29,999 HKD ($3840)	11.7 %	24.3 %	11 %
13. 30,000–39,999 HKD ($5121)	4.5 %	13.5 %	12 %
14. 40,000–49,999 HKD ($6401)	3.6 %	8.1 %	6 %
15. 50,000–59,999 HKD ($7681)	6.3 %	2.7 %	10 %
16. 60,000–69,999 HKD ($8961)	1 %	4.5 %	7 %
17. 70,000–79,999 HKD ($10,242)	1 %	2.7 %	5 %
18. 80,000–89,999 HKD ($11,522)	0.0 %	1.8 %	7 %
19. 90,000–99,999 HKD ($12,802)	2.7 %	1.8 %	4 %
20.>100,000HKD	1.8 %	1.8 %	12 %
Parental education (scored below from 1 to 10)			
1–4.Did not graduate from secondary school	37.6 %	39.2 %	
5.Graduated from Secondary school	6.4 %	6.9 %	
6.Associate degree or higher diploma	20.2 %	15.7 %	
7.Bachelor degree	22 %	22.5 %	
8.Graduate degree	0 %	1 %	
9.Master degree	13.8 %	11.8 %	
10.Dotoral degree	0 %	2.9 %	

Note. The information was gathered from parents of 111 (79 %) participating children.

2.3. Measures

A battery of eight tests were administered to the participating children. The raw score sums were calculated for all variables and used in the analyses. The Cronbach's alpha was employed to indicate the reliability of all the measures in this study.

2.3.1. Visual pattern understanding

This was measured with three types of pattern problems: repeating patterns, growing patterns, and rotation patterns. *The repeating pattern task* is a 10-item task developed by Rittle-Johnson et al. (2013) that tests the ability to duplicate (1 item), extend (1 item), and abstract repeating patterns (6 items) and identify the unit of repeat (2 items). In each trial, an experimenter created a model pattern with 2 units in front of the child (see Fig. 1A for a sample item). In the "duplicate" condition, a child was asked: "Make the same pattern as mine, with the same number of blocks in the same places as mine.'' In the "extend" condition, a child was asked: "Make the same pattern as mine, and then continue using more blocks." In the "abstract" condition, a child was shown a pattern and was asked to create the same pattern using blocks of different colors and shapes. In the "unit identification" condition, children were shown an AAB pattern and were asked: "Can you move the stick to show where the pattern starts over again?

The visual growing pattern task is a 10-item task developed by the corresponding author (Chen et al., 2024) and modeled on Papi et al. (2011) that tests the ability to extend visual growing patterns. In each item, the experimenter showed the child a model pattern with three units (see Fig. 1B for a sample item). The child was asked: "Here is a pattern I made. Can you continue using these dots?" Children were asked to create the next unit. The arrays in a unit were increasing in five items and were decreasing in the other five. The unit of increasing (decreasing) was 1 for item one (six), 2 for item two (seven), 3 for item three (eight). Item four (nine) and five were triangular and square patterns, with the varying unit of change with position. Item ten was composed of two decreasing sub-patterns with the unit being one and two respectively. The rationale for adding the complex patterns (triangle and square patterns) is that understanding of spatial structural patterns other than arrays is one goal of mathematics education, and needs to be assessed in growing pattern understanding. Additionally, the complex patterns also allow us to increase the range of the difficulty level of the ten items so that the growing pattern task can be used in a longitudinal project to capture the growing pattern skills of children at different ages.

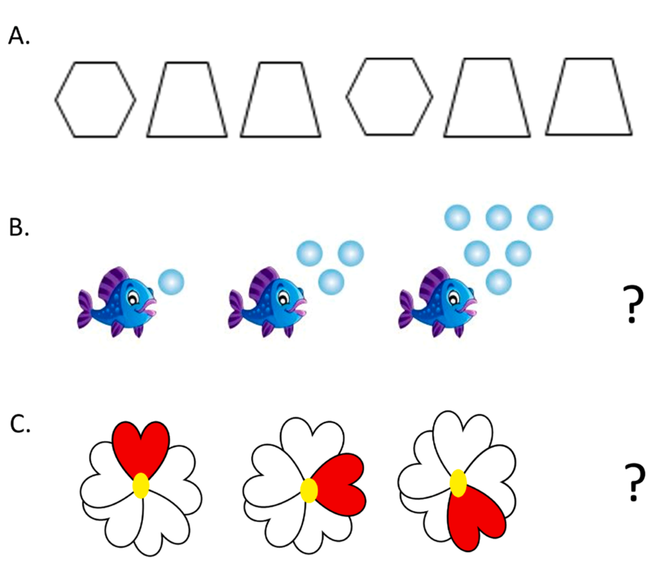

Fig. 1. Sample items for the (A) repeating pattern task, (B) growing pattern task, (C) rotating pattern task.

Children were told to create rather than choose from options, because generative tasks are free of the bias of guessing and open the possibility of analyzing strategies in the future. Moreover, the repeating pattern used the generative response. We wanted to make sure that the type of response is aligned across tasks.

The rotation pattern task is a 10-item task developed by the corresponding author and modeled on Kidd et al. (2014) that tests the ability to extend rotation patterns. The 10 items featured patterns varying in rotating direction (clockwise and counter-clockwise) and degree of rotation per unit (90° and 180°). In each item, the experimenter showed the child a pattern of three units and asked the child to create the fourth unit. To adapt the test to the ability level of the young participants and avoid the floor effect, we added reference lines to a few items to mark the change in position of the elements within a pattern (see Fig. 1C for a sample item). The reliability for the three tasks of pattern understanding was 0.82.

2.3.2. Working memory

This was measured using the Wechsler Intelligence Scale for Children–III backward digit span task (Wechsler, 1991). The test requested participants to recall a sequence of spoken digits in the reverse order. Test trials began with two numbers and increased by one number in each block, which contained three sequences. The test was halted if a child recalled all the three sequences wrongly in one block. The number of sequences recalled correctly backward determined the working memory score. The reliability coefficient was 0.79.

2.3.3. Spatial visualization

Spatial visualization was measured with the subtest of spatial relations from the Woodcock-Johnson Tests of Cognitive Abilities (2001). The participating children were asked to identify, from among distractors a set of pieces constituting a complete shape presented as a target (i.e., "Two of these pieces go together to make this. Tell me which two pieces."). The test was stopped after three minutes. This task involves spatial reasoning and problem solving (Linn & Petersen, 1985), which are essential components of fluid intelligence (Cattell, 1963). The reliability coefficient was 0.79.

2.3.4. Vocabulary

The measure of receptive vocabulary was a shortened version (Chow et al., 2008) of the Hong Kong Cantonese Receptive Vocabulary Test (Cheung et al., 1997). The test had 31 items. For each item, a child heard a word and was shown four pictures. Children responded by selecting the picture that matched the spoken word from three distractors. The reliability coefficient was 0.70.

2.3.5. Phonological awareness

Phonological awareness was measured by the syllable deletion task consisting of 16 three-syllable phrases. Each three-syllable phrase was presented orally by a tester, and the child was asked to say the remainder of the phrase after deleting one syllable. For example, for the phrase / din6 si6 gei1 / (電視機, television), with the Arabic numerals indicating Cantonese lexical tones, the child was asked to say what was left without /si6/, with the correct answer being/ din6 gei1 /. The test has been used in previous studies with Hong Kong children (e.g., Zhang & Lin, 2018). The reliability coefficient was 0.94.

2.3.6. Morphological awareness

Morphological awareness was measured using the Morphological Construction Test (McBride-Chang et al., 2007), which taps into children's ability to invent novel compound words by combining acquired morphemes. Each three-sentence story scenario was read to the child. For example, one scenario is "When a type of oil is made from peanut, we call that peanut oil. What should we call the type of oil that is made from mushroom?" The correct answer is "mushroom oil." One point was given for each correct answer. The task contains 27 items. The reliability coefficient was 0.92.

2.3.7. Orthographic awareness

Orthographic awareness was measured by the character decision task. The child was asked to indicate whether a presented stimulus was a real Chinese character or not. The 50 items were presented in random orders and were borrowed from Tong et al. (2009), with 10 drawings, 10 pseudo-characters and 30 real characters as fillers. Only responses to the drawing and pseudo-character items were scored. Following Qian et al. (2015), responding "Yes" to the pseudoword items was coded as "correct" because they conformed to the orthographic rules. The reliability coefficient was 0.86.

2.3.8. Chinese word reading abilities

Two kinds of word reading abilities were measured at the second time point, i.e., single-character word reading and double-character word reading. Each task had 60 items arranged in the ascending order of difficulty based on Hong Kong kindergarten and primary school textbooks. The test was halted when the child had failed to read 10 consecutive words. The task has been used for Hong Kong children in previous studies (Zhang & Lin, 2018). The reliability coefficients were 0.94 for single-character and 0.95 for double-character word reading.

2.4. Data analysis

To answer the first question, a multivariate multiple regression (MMR) was conducted with Chinese single-character word reading and double-character word reading as two dependent variables. The predictors included pattern understanding, working memory, spatial visualization, vocabulary, family income, and age. The covariances between the two dependent variables and between the predictors were also estimated. The MMR model was saturated; thus, goodness-of-fit indices were perfect and cannot be used to evaluate the model-data fitness. Nonetheless, it is possible to proceed with the interpretation of parameters (Muthén & Muthén, 2017).

To answer the second question, two models, i.e. the full model and indirect effect model, were compared to select the path model which predicts Chinese single- and double-character word reading simultaneously. The full model hypothesized that pattern understanding had direct effects on word reading skills beyond the indirect effects via the three metalinguistic awareness skills, which were added as predictors of T2 word reading skills. The two direct-effect paths were added in addition to predictive paths from other T1 skills, income, and age. Also included were the paths from T1 cognitive-and-linguistic skills to T1 metalinguistic awareness skills. The indirect effects of T1 pattern understanding on T2 word reading skills via T1 metalinguistic awareness skills were estimated. The covariances between the two dependent variables, between the predictors, and between the mediators were also estimated. The indirect effect model hypothesized that the effects of pattern understanding were fully mediated by the metalinguistic awareness skills, and thus did not include the two paths from T1 pattern understanding to T2 word reading skills. Full information maximum likelihood was adopted to estimate model parameters for both the MMR and path models. This method permits missing data without list- or pairwise deletion of incomplete cases. Thus, all 138 children were included in the analysis. Chi-squared test was performed for model comparison. The analyses were performed on the R platform using the lavaan package (Rosseel, 2012). The model fit indices and parameters of the full and indirect models are presented in the supplementary materials.

3. Results

The means, ranges, standard deviations, and distribution characteristics for all study variables are shown in Table 2. The correlation coefficients between them are shown in Table 3. Pattern understanding at T1 were correlated with T2 single-character (r = 0.38) and double-character word reading (r = 0.47), both of medium strength (Cohen, 1998). T2 word reading was also significantly correlated with the three metalinguistic awareness skills, specifically with OA of small strength (single-character: r = 0.22), with MA of medium strength (single-character: r = 0.42; double-character: r = 0.50), and with PA of large strength (single-character: r = 0.57, double-character: 0.59). Additionally, the PA (r = 0.49) and MA (r = 0.37) were also associated with pattern understanding, both of medium strength.

Table 2
Descriptive statistics of all the measures in the study.

	n	mean	sd	min	max	skew	kurtosis
Age	111	59.60	5.48	51.00	79.00	0.76	0.65
Family income	111	12.04	3.64	1.00	20.00	−1.11	2.31
t2 DCWR	129	13.10	10.37	0.00	49.00	1.19	1.28
t2 SCWR	129	19.50	11.53	0.00	50.00	0.75	−0.39
t1 Pattern	131	12.79	5.02	1.00	26.00	0.01	−0.59
t1working memory	137	2.26	2.17	0.00	11.00	0.55	0.13
t1SV	136	10.53	3.36	0.00	19.00	−1.06	1.91
t1vocabulary	137	19.01	3.76	7.00	26.00	−0.38	−0.2
t1PA	138	6.44	5.46	0.00	16.00	0.31	−1.32
t1MA	136	8.69	6.00	0.00	25.00	0.72	0.10
t1OA	134	14.5	4.53	0.00	20.00	−0.38	0.78

Note. SCWR, Single-character word reading; DCWR, double-character word reading; SV, spatial visualization; PA, phonological awareness; MA, morphological awareness; OA, orthographic awareness; WM, working memory.

3.1. Does pattern understanding have a unique effect on word reading independent of domain-general cognitive and language skills and demographic information?

The regression coefficients estimated in the MMR model are displayed in Table 4. For T2 single-character word reading, T1 pattern understanding, vocabulary, and working memory were the significant predictors (all $p < 0.05$). Spatial visualization, age, and family income did not significantly predict T2 single-character word reading (all $p >$ 0.05). The results are similar for the prediction of T2 double-character word reading. T1 pattern understanding, vocabulary, and working memory were the significant predictors (all $p < 0.05$), whereas spatial visualization, age, and family income were not (all $p > 0.05$). The model explained 27 % of the variance in single-character Chinese word reading, 36 % of the variance in double-character Chinese word reading at T2.

3.2. Do metalinguistic awareness skills mediate the relation of pattern understanding to word reading?

The result of the model comparison indicated that the full model fit the data significantly better than the indirect effect model ($\Delta\chi^2 = 6.43$, $\Delta df = 2$, $p = .04$). Therefore, the direct effect model was interpreted. The full model, as shown in Fig. 2, had a good fit (fit indices: CFI = 1.000; TLI = 1.043; RMSEA = 0.001; SRMR = 0.019, $\chi^2 = 3.49$, df = 6, $p =$ .745). For T2 double-character word reading, the path from pattern understanding was significant (standardized estimate = 0.26, $p < .05$), indicating a significant direct effect. Pattern understanding also had a significant indirect effect on double-character word reading via PA (estimate = 0.07; 95 % confidence interval: [0.001, 0.146]). Neither MA nor OA significantly predicted T2 double-character word reading (both $p > .05$). The two skills also showed no significant mediation of the association between pattern understanding and double-character word reading (MA: estimate = 0.021; 95 % confidence interval: [-0.015, 0.056]; OA: estimate = 0.004; 95 % confidence interval: [-0.015, 0.023]). The model explained 47 % of the variance in T2 double-character Chinese word reading.

For T2 single-character word reading, the path from pattern understanding was nonsignificant (standardized estimate = 0.18, $p = .099$).

Table 3
Correlation coefficients between measures.

	2	3	4	5	6	7	8	9	10	11
1. Age	−0.20	.16*	.05	−0.05	.12	.08	.18	.01	.03	.05
2. Income		−0.03	.13	.07	−0.01	.10	−0.03	.05	.10	.11
3. Pattern			.49**	.37**	.18	.56**	.54**	.36**	.38**	.47**
4. PA				.54**	.22*	.51**	.34**	.39**	.59**	.57**
5. MA					.22*	.37**	.18*	.57**	.50**	.42**
6. OA						.13	.23**	.35*	.22*	.19
7. WM							.36**	.29*	.40**	.36**
8. SV								.27*	.23*	.19*
9. Voc									.37**	.44**
10. SCWR										.91**
11. DCWR										1

Note. SV, spatial visualization; PA, phonological awareness; MA, morphological awareness; OA, orthographic awareness; WM, working memory; SCWR, Single-character word reading; DCWR, double-character word reading

Table 4
Regression coefficients predicting T2 word reading abilities from demographic variables and T1 abilities.

	Double-character word reading			Single-character word reading		
	Est	*se*	*p*	*Est*	*se*	*p*
Age	0.001	0.10	.99	0.05	0.10	.65
Income	0.09	0.09	.31	0.10	0.10	.29
t1pattern	0.35	0.11	.001	0.27	0.12	.02
t1vocabulary	0.27	0.08	.001	0.23	0.09	.01
t1working memory	0.18	0.08	.03	0.20	0.09	.02
t1spatial visualization	−0.06	0.09	.52	−0.07	0.10	.51

Pattern understanding had a significant indirect effect on single-character word reading via phonological awareness (estimate = 0.084; 95 % confidence interval: [0.001, 0.146]). Neither MA nor OA significantly predicted T2 single-character word reading (both $p > .05$). The two skills showed no significant mediation of the association between pattern understanding and single-character word reading (MA: estimate = 0.016; 95 % confidence interval: [-0.018, 0.049]; OA: estimate = 0.004; 95 % confidence interval: [-0.016, 0.024]). The model explained 39 % of the variance in T2 single-character Chinese word reading.

In addition, T1 working memory significantly predicted PA (standardized estimate = 0.33, $p < .001$) and MA (standardized estimate = 0.20, $p < .05$); T1 vocabulary significantly predicted PA (standardized estimate = 0.20, $p < .05$), MA (standardized estimate = 0.48, $p < .001$) and OA (standardized estimate = 0.34, $p < .001$).

4. Discussion

The present study investigated the direct and indirect contributions of pattern understanding to Chinese word reading abilities. The results showed that pattern understanding predicted word reading a year later, beyond the effects of working memory, spatial visualization, receptive vocabulary and demographic variables. The predictive effect was partially mediated by phonological awareness for double-character word reading, while fully mediated by phonological awareness for single-character word reading. The findings speak to the importance of pattern understanding for young children in learning to read in Chinese.

4.1. The unique effect of pattern understanding on Chinese word reading

Pattern understanding made a unique contribution to word reading abilities 14 months later independently of spatial visualization, working memory, vocabulary and demographic variables. The unique

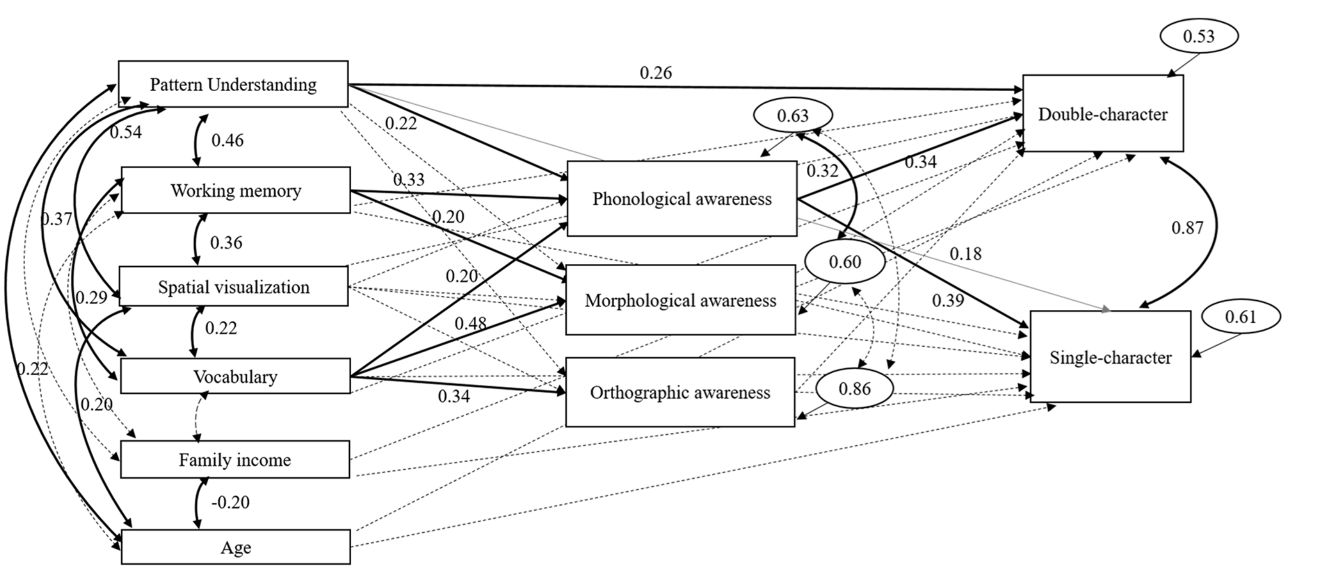

Fig. 2. The standardized solution of the full model containing direct and indirect effects of patterning on word reading controlling for working memory, fluid intelligence, vocabulary, and background information.
Note. The solid lines represent the significant coefficients ($p < 0.05$) and the dashed lines represent the non-significant coefficients ($p > 0.1$). The gray line represents the coefficient at the marginal level of statistical significance ($0.05 < p < .1$). The lines with double-headed arrow indicate covariance, and the lines with single-headed arrow indicate predictive paths.

contribution falls in line with previous findings on the importance of visual pattern understanding in word reading in alphabetic languages (Burgoyne et al., 2019). It is one of the first findings to demonstrate its importance in Chinese word reading beyond other domain general skills. Previous studies have revealed that number pattern understanding contributes to Chinese word reading (Liu et al., 2022). The present finding extended previous findings by excluding the confounding factor of symbolic understanding and establishing the longitudinal prediction of earlier visual pattern understanding to subsequent Chinese word reading. In addition to pattern understanding, working memory and vocabulary each made a unique contribution to word reading abilities 14 months later after demographic variables were controlled for. The current findings add to the literature about the important roles that working memory and vocabulary in learning to read in young Chinese children (Chung & McBride-Chang, 2011; Huo et al., 2021).

Previous studies have shown the importance of fluid intelligence in word reading, especially in kindergarten children (Guerin et al., 2020; Yang et al., 2019). The present study contributed to this area by tentatively delineating the contribution of two components in fluid intelligence, i.e., rule abstraction (indexed by pattern understanding) and spatial visualization. The findings showed that visual pattern understanding significantly predicted subsequent word reading, but spatial visualization did not, suggesting that it might be the process of rule abstraction that drives the association between fluid reasoning and word reading, not spatial visualization. Some previous findings have shown that spatial visualization measured by the same task as the present study, predicted Chinese word reading in addition to demographic variables and vocabulary (Yang et al., 2021). Such predictive effect was not found in the current study. The child participants were 1 year older in our study. It is possible that the role played by visual spatial skills decreases as children become more experienced with reading (Luo et al., 2013). The alternative explanation is that spatial visualization exerts an effect on the growth of Chinese reading in kindergarten children rather than the concurrent baseline as revealed in Yang et al. (2021). However, the present study cannot tease apart these explanations as the baseline reading performance was not assessed. The limitation was discussed in the later section.

In light of previous findings that earlier pattern understanding predicted subsequent calculation skills in kindergarten and grade 1 children (Burgoyne et al., 2019; Pasnak et al., 2015, 2016), the present findings on its contribution to word reading abilities suggest that pattern understanding is a domain-general skill that plays roles in various academic areas. Apart from its direct contribution to academic outcomes, pattern understanding may also exert an indirect effect though domain specific path. For example, pattern understanding taps into relational thinking (Miller et al., 2016), which contributes to understanding the relation between items in arithmetic equations. For word reading abilities, pattern understanding may contribute to the abstractive and segmental representation of speech, which sets the stage for the speech-print mapping in learning to read. This process is discussed in detail below.

4.2. The mediating roles of metalinguistic awareness in the relation between pattern understanding and Chinese word reading

Our hypothesis that metalinguistic awareness skills mediate the influence of visual pattern understanding on word reading is partially confirmed. Phonological awareness mediated the role of pattern understanding in single- and double-character word reading. This mediating pathway confirmed the hypothesis that pattern understanding enhances word reading through facilitating phonological skills. Phonological awareness, the ability to segment speech into units, is the critical component in understanding the sound-print mapping relationships. Phonological awareness at the syllable level has been shown repeatedly to play a substantial role in character and Chinese word reading in kindergarten children (e.g. Zhang & Lin, 2018). Moreover, the processes of rule abstraction indexed by visual pattern understanding is likely to be involved in phonological awareness; abstractive thinking enables one to detach from the meaning of the language and think about the phonological structure of the speech (Kleeck, 1982; Tunmer et al., 1988). A good representation of syllables facilitates segmenting speech and mapping it onto print. In the meantime, we acknowledge that there is a possibility that the relation between pattern understanding and phonological awareness could be driven by a third variable, e.g., listening comprehension, which was not measured in this study.

In addition to pattern understanding, working memory and vocabulary concurrently predicted phonological awareness. This finding adds to the literature that the development of phonological awareness relies on cognitive growth (e.g., Tunmer et al., 1988) and vocabulary (e.g. Metsala & Walley, 1998). In summary, the role phonological awareness plays in mediating the association between domain-general abilities and word reading converges with findings in the alphabetic context (e.g. Hulme et al., 2015), which point to a developmental pathway that is culturally universal.

Our hypothesis that morphological awareness would mediate the relation between pattern understanding and word reading was not confirmed. There are several explanations for this. The first might be a suppression effect due to large covariance between phonological awareness and morphological awareness. After removing phonological awareness from the model, morphological awareness did indeed significantly predict subsequent word reading abilities ($\beta = 0.23$, $p = .03$). However, pattern understanding still did not make a unique contribution to MA after removing PA. This tentatively speaks to the explanation that morphological awareness requires little abstractive thinking, which is the essence of visual pattern understanding. The compounding morphology is inherently about word meaning, in contrast to phonology which can be separated from meaning. The development of morphological awareness barely requires detaching from meaning of language, and thus relies less on abstractive thinking. Admittedly, morphological awareness requires multi-step analogical reasoning (Anglin et al., 1993), which relies on working memory. The contribution of working memory and vocabulary to morphological awareness was corroborated by the significant paths found in the current study, which is consistent with previous studies in Chinese preschool children (Huo et al., 2021).

Finally, the mediating role of orthographic awareness was not confirmed either. Orthographic awareness was shown to be weakly but significantly correlated with single-character word reading. The association is consistent with previous findings (Huo et al., 2021; Yang et al., 2019). The small effect size did not survive other skills predicting word reading. None of the domain general cognitive abilities, i.e., pattern understanding, working memory, or spatial visualization, contributed to orthographic awareness beyond vocabulary and demographic information. The character-decision task of orthographic awareness focuses on distinguishing between Chinese characters and drawing and measures preliminary knowledge about the Chinese writing system, and this may rely on exposure more than inductive thinking (Levy et al., 2006). Instead, vocabulary made a significant contribution to orthographic awareness. Together with previous findings (Huo et al., 2021), it is suggested here that a basic level of vocabulary knowledge, possibly serving as a proxy for general language development, is the foundation for metalinguistic awareness skills and word reading in young children.

After the three metalinguistic awareness skills were considered, the direct effect of pattern understanding on double-character word reading was still significant. This finding is somewhat surprising, as the metalinguistic awareness skills have been established to be robust precursors of Chinese word reading and account for a considerable amount of individual differences in previous studies (e.g., Tong et al., 2009). On the other hand, the direct effect of visual pattern understanding was not significant on single-character word reading, speaking to the variation of the role pattern understanding plays across types of words.

The contribution that visual pattern understanding makes to double-character word reading might be due to its overlap with analogical reasoning, which also touches on inductive and deductive reasoning (Goswami, 2011). Extracting sub-lexical orthographic rules (letter patterns and stroke patterns in the case of Chinese characters) and applying them when encountering a novel word, termed as a self-teaching mechanism, is shown to be an important process in reading acquisition (Goswami & Mead, 1992). The self-teaching process could also happen at the lexical (character) level (Li et al., 2014), which refers to the use of knowledge about characters to infer pronunciation and meaning of double-character words. This lexical level self-teaching process might underlie double-character word reading, but not single character reading in kindergarten children. Self-teaching mechanism for character reading requires sub-lexical orthographic knowledge (e.g. radical position and function knowledge), which is still limited for kindergarten children (Ho et al., 2003). The self-teaching process is based on analogical thinking (Ho et al., 1999; Share, 1999). However, analogical reasoning was not assessed in the present study, which prevents us from excluding or confirming this possibility.

The present finding suggests that visual pattern understanding makes a unique contribution to learning to read Chinese words independently of working memory, spatial visualization, and vocabulary. Pattern understanding, which is characterized by rule abstraction (Pasnak et al., 2015; Zippert et al., 2019), contributes to phonological awareness at the syllable level in young children, which is important for both single- and double-character word reading. Moreover, pattern understanding has a direct effect on double-character word reading beyond metalinguistic awareness skills, indicating the cognitive differences between single- and double-character reading in young children.

4.3. Limitations

Although the present study provides novel findings regarding the role of visual patterning understanding in Chinese word reading, it presents several limitations. First, children's initial level of Chinese word reading was not controlled for. Thus, inference regarding influence on the longitudinal change of word reading ability cannot be made. Second, we did not examine pattern understanding using sequences of Chinese characters which reflect orthographic rules in the writing system. It is possible that visual pattern understanding exerts an influence on Chinese word reading via the domain specific pattern understanding. Future studies could examine this possibility. Third, the present longitudinal study was a short-term one with a one-year interval. The mechanism of how pattern understanding influences word reading abilities may change as children grow. For example, children develop sensitivity to sub-lexical orthographic patterns in primary school years. At that time, pattern understanding may contribute to word reading through an orthographic learning mechanism, i.e., to identify new characters/words based on their orthographic knowledge (Ho et al., 1999). At last, the current study is correlational in nature, and thus causal claims cannot be made.

4.4. Educational implications

The present study suggests that pattern understanding plays a role in Chinese word reading abilities. One path that pattern understanding exerts an effect is via phonological awareness, which is a core precursor of word reading. Phonological awareness is one of the critical components in early literacy education programs in alphabetic and logographic contexts. The phonological skill-based instruction has shown to be effective in both contexts (English: Müller et al., 2020; Chinese: Li et al., 2020). Our findings suggest that including patterning activities might enhance the effects of phonological instructions on word reading. It is also possible that children with good pattern understanding might benefit more from phonological instructions. Moreover, pattern understanding directly contributes to double-character word reading but not single-character word reading. This finding together with previous studies suggests that slightly different teaching strategies may be needed to teach these two types of word reading. The present finding highlights the role of pattern understanding in double-character word reading in children as young as five and six years old. Teachers may consider creating sequences of double-character words with underlying rules, e.g. containing shared morpheme. Admittedly, these suggestions are only tentative as we only provide correlational evidence. Intervention studies are needed in the future.

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Declaration of competing interest

We have no known conflict of interest to disclose.

Data availability

Data will be made available on request.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ecresq.2023.12.016.

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